

MOHAMMAD HOSSEIN MOHAMMADI

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Research Interests

- Large Language Models (LLMs) & Generative AI
- Computer Vision & Medical Image Analysis
- Human-Centered AI & Intelligent Systems Design
- Explainable AI (XAI) & Trustworthy Machine Learning

Education

Islamic Azad University, Najafabad Branch

Sep. 2015 – Sep. 2021

Bachelor of Science in Computer Engineering (Information Technology)

Najafabad, Iran

- **Thesis Title:** Semantic segmentation of breast cancer pathology images using the **U-Net** model.
- **Supervisor:** Dr. Mehdi Jabalameli

Patents

- [1] **The system for early detection of Heterochromia disease in the neonates**
Seyed Yahya Moradi, [M.H. Mohammadi](#)
Patent No: IR-139850140003011312, Publication Date: 2021
- [2] **Pocket Protector Of Sensitive Injective And Inhalant Vials Against Temperature And Impact Changes**
Seyed Yahya Moradi, [M.H. Mohammadi](#), Pejman Mohammadi
Patent No: IR-139850140003011419, Publication Date: 2020

Books

- [1] **A Step-By-Step Guide to Patenting in Iran** (View on Amazon)
[M.H. Mohammadi](#), Ali Karimi, Seyed Yahya Moradi
Publisher: Arna, ISBN: 978-600-356-922-8, Language: English, Year: 2020
- [2] **Basics of Artificial Intelligence**
[M.H. Mohammadi](#), Seyed Yahya Moradi, Mahsa Mohammadi
Publisher: Arna, ISBN: 978-622-291-016-7, Language: Persian, Year: 2021
- [3] **Basics of Internet of Things (IoT)**
Seyed Yahya Moradi, [M.H. Mohammadi](#)
Publisher: Arna, ISBN: 978-622-2910-14-3, Language: Persian, Year: 2021

Projects

Multimodal AI: Image Caption Generation System | *Python, CNN, LSTM, NLP, Computer Vision* May 2026

- Developed an end-to-end AI system at the intersection of Computer Vision and NLP to generate contextually relevant textual descriptions for unstructured image data.
- Architected a hybrid neural network utilizing Convolutional Neural Networks (CNN) for robust feature extraction and Long Short-Term Memory (LSTM) networks for accurate sequence generation.
- Trained, optimized, and validated the deep learning model on the Flickr8k dataset, managing complex data preprocessing pipelines for sequential text tokens and images.
- Focused on writing clean, modular, and object-oriented Python code to ensure high system maintainability and scalability.

End-to-End Speech Recognition Web Service | *Python, Deep Learning, Streamlit, Git, UI/UX* Sep. 2025

- Designed, trained, and deployed an automated Speech Recognition system to process sequential audio streams and map extracted acoustic features to text using deep neural networks.
- Built a robust data preprocessing pipeline to handle raw audio signals and bridge the gap between machine learning performance and user experience.
- Engineered a human-centered, interactive web dashboard using Streamlit, allowing end-users to seamlessly record audio, upload files, and view real-time transcriptions.
- Applied UX principles and modular code architecture (with Git version control) to ensure visual consistency, accessibility, and readiness for CI/CD integration.

HealthTech AI: Medical Image Segmentation | *PyTorch, U-Net, FastAPI, Docker, UI/UX* Jan. 2025

- Engineered a production-ready deep learning application for the semantic segmentation of breast cancer pathology images, transforming academic research into a scalable software product.
- Developed and fine-tuned a U-Net architecture to precisely delineate tissue boundaries and detect tumor regions, maximizing diagnostic potential.
- Packaged the trained PyTorch model into a high-performance RESTful API using FastAPI, and containerized the backend using Docker for isolated, cloud-ready deployment.
- Conceptualized an intuitive frontend interface leveraging Product Design principles, enabling physicians to upload scans and analyze segmentation masks without technical friction.

Work Experience

Product Designer (Design System Specialist)

Sep. 2025 – Present

TeamNey

Scotland, United Kingdom (Remote)

- Architected and maintained a robust Design System, establishing reusable UI components, standardized typography, and unified design tokens to significantly accelerate the frontend development cycle.
- Bridged the gap between design and software engineering teams, ensuring pixel-perfect and technically feasible implementation of interactive prototypes.
- Conducted comprehensive UX audits and enhanced cross-platform accessibility, ensuring the final digital products were highly intuitive and user-centric.
- Thrived in a remote, international work environment, effectively communicating complex design architectures and user journey strategies to cross-functional stakeholders.

UI/UX Designer

Feb. 2022 – Sep. 2023

PizzaBar

Copenhagen, Denmark

- Designed and delivered a custom CRM solution using user-centered design principles to improve operational efficiency and user satisfaction.
- Led end-to-end design from research and prototyping to system deployment, ensuring scalability and usability.
- Developed a scalable design system and interactive prototypes in Figma, enhancing consistency and collaboration across teams.
- Conducted user research and usability testing to align business goals with user needs, resulting in a polished, high-impact product.

Honors & Awards

2nd Place Winner, 7th Mobile Programming Marathon | Sharif University of Technology

Sep. 2019

- Achieved 2nd place out of 85 competing teams from across Iran in a highly competitive, time-bound software development hackathon.
- Demonstrated rapid problem-solving, algorithmic thinking, and effective teamwork under strict deadlines to build functional software solutions as a core member of the “Chelesme” team.

3rd Place Winner, 4th Mobile Programming Marathon | Sharif University of Technology

Sep. 2017

- Secured 3rd place out of 90 competing teams nationwide in an intensive software development hackathon.
- Delivered functional applications through rapid prototyping, algorithmic problem-solving, and robust coding under strict deadlines alongside the “Yakhmak” team.

Skills

Languages & Computing: Python, C++, Java, JavaScript, SQL, NumPy, Pandas
LLMs & Generative AI: Transformers, LLM Fine-Tuning, RAG, LangChain, Hugging Face, Prompt Engineering, Diffusion Models
Computer Vision & DL: PyTorch, OpenCV, Medical Image Analysis, Image Segmentation (U-Net), ViT, XAI (SHAP, LIME)
AI Infrastructure & MLOps: Docker, FastAPI, MLflow, Git, GitHub Actions CI/CD, AWS
HCI & Intelligent Systems: Figma, Design Systems, Streamlit, Interactive Prototyping, Human-in-the-Loop AI

Languages

- English (B2)
- Persian (Native proficiency)

Professional References

References available upon request.